

WEEK 1 ASSIGNMENT REPORT BY BHARATH KOWSHIK ULLANGULA (24JR1A05C5)
AI Foundations & Developer Environment

Q1. What is Artificial Intelligence?

Artificial Intelligence (AI) is a branch of computer science that enables machines to perform tasks that normally require human intelligence. These tasks include learning, reasoning, problem-solving, understanding language, recognizing images, and making decisions. AI systems use data and algorithms to improve their performance over time.

Q2. Difference Between AI, ML, Generative AI and Agentic AI

Technology	Description
AI	Broad field of creating intelligent machines
Machine Learning	Subset of AI that learns from data
Generative AI	Creates new content such as text, images, audio, and code
Agentic AI	AI systems that can plan, reason, and take actions autonomously

Q3. Explain any three open-source AI models available on Hugging Face.

➔ **lama 3**

- Developed by Meta
- Open-source large language model
- Used for chatbots and content generation

➔ **Mistral 7B**

- Developed by Mistral AI
- Lightweight and efficient LLM
- Suitable for local deployment

➔ **FLAN-T5**

- Developed by Google
- Instruction-tuned language model
- Performs well on question answering tasks

Q4 . What is the difference between traditional programming and AI-assisted development?

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Traditional Programming AI-Assisted Development

Developer writes all logic AI suggests code

Rule-based

Data-driven

Fixed outputs

Adaptive outputs

Slower development

Faster development

Q5. Explain AI hallucinations with one example.

AI hallucination occurs when an AI model generates incorrect or fabricated information while presenting it confidently.

Example:

If an AI says a programming language was invented in 2028 even though it doesn't exist, that is a hallucination.



FIGURE : AI GENERATED AS I ASK TO MAKE AN IMAGE AS A G WAGON CAR FLYING IN AIR